

Day 19 - Lecture 19.5: Advanced Operations on Box Plots

1. Overview & Purpose

Standard box plots can be significantly customized to provide richer statistical insight and better presentation aesthetics. This lecture covers:

- **Horizontal Box Plots** (`vert=False`)
 - **Displaying the Mean Marker** (`showmeans=True` , `meanline=True`)
 - **Customizing Whiskers** (`whis` parameter)
 - **Comparing Multiple Groups** with custom labels
-

2. Advanced Parameters of `plt.boxplot()`

```
plt.boxplot(  
    x,                    # Dataset or list of datasets: [group1, group2, ...]  
    vert=True,           # If False, plots horizontally (useful when category  
    showmeans=True,      # If True, plots a marker for the mean  
    meanline=True,       # If True with showmeans=True, renders mean as a dotted  
    whis=1.5,            # Whisker multiplier (e.g. 1.5, 2.0, or percentiles [0.01, 0.99]  
    patch_artist=True,    # If True, fills boxes with color (allows facecolor & edgecolor  
    notch=False,         # If True, creates notched box to visually test median equality  
    tick_labels=['A', 'B'] # Labels for each dataset (formerly `labels`)  
)
```

3. Comparing Mean vs. Median in Box Plots

When `showmeans=True` :

- The **solid line** inside the box is the **Median** (Q_2).
- The **green triangle** or **dashed line** is the **Arithmetic Mean** (μ).

- If Mean $\approx$ Median: Distribution is symmetric.
- If Mean is noticeably to the right (higher) than Median: Right-skewed (driven by high outliers).

4. Code Implementation

```
import matplotlib.pyplot as plt
import numpy as np

# 1. Horizontal Box Plot with Mean line
data = [7, 8, 5, 6, 9, 4, 10, 12, 15]

plt.figure(figsize=(8, 4))
plt.boxplot(data, vert=False, showmeans=True, meanline=True, whis=2.0)
plt.title("Horizontal Box Plot with Mean Line & Extended Whiskers (whis=2.0)", fontweight="bold")
plt.xlabel("Values", fontsize=11)
plt.grid(True, linestyle=":", alpha=0.6)
plt.show()

# 2. Comparing Multiple Datasets
np.random.seed(42)
group1 = np.random.normal(50, 10, 100)
group2 = np.random.normal(60, 15, 100)

plt.figure(figsize=(8, 5))
box = plt.boxplot([group1, group2], tick_labels=["Group 1", "Group 2"], patch_artist=True)

# Styling individual boxes
colors = ['lightsteelblue', 'lightcoral']
for patch, color in zip(box['boxes'], colors):
    patch.set_facecolor(color)

plt.title("Comparison Between Two Groups", fontsize=14, fontweight="bold")
plt.ylabel("Score / Performance Metric", fontsize=11)
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.show()
```

5. Key Takeaways

- `vert=False` is especially helpful when dealing with lengthy category names to prevent label overlap.

- Setting `patch_artist=True` converts the wireframe box into a filled polygon, enabling company branding colors.